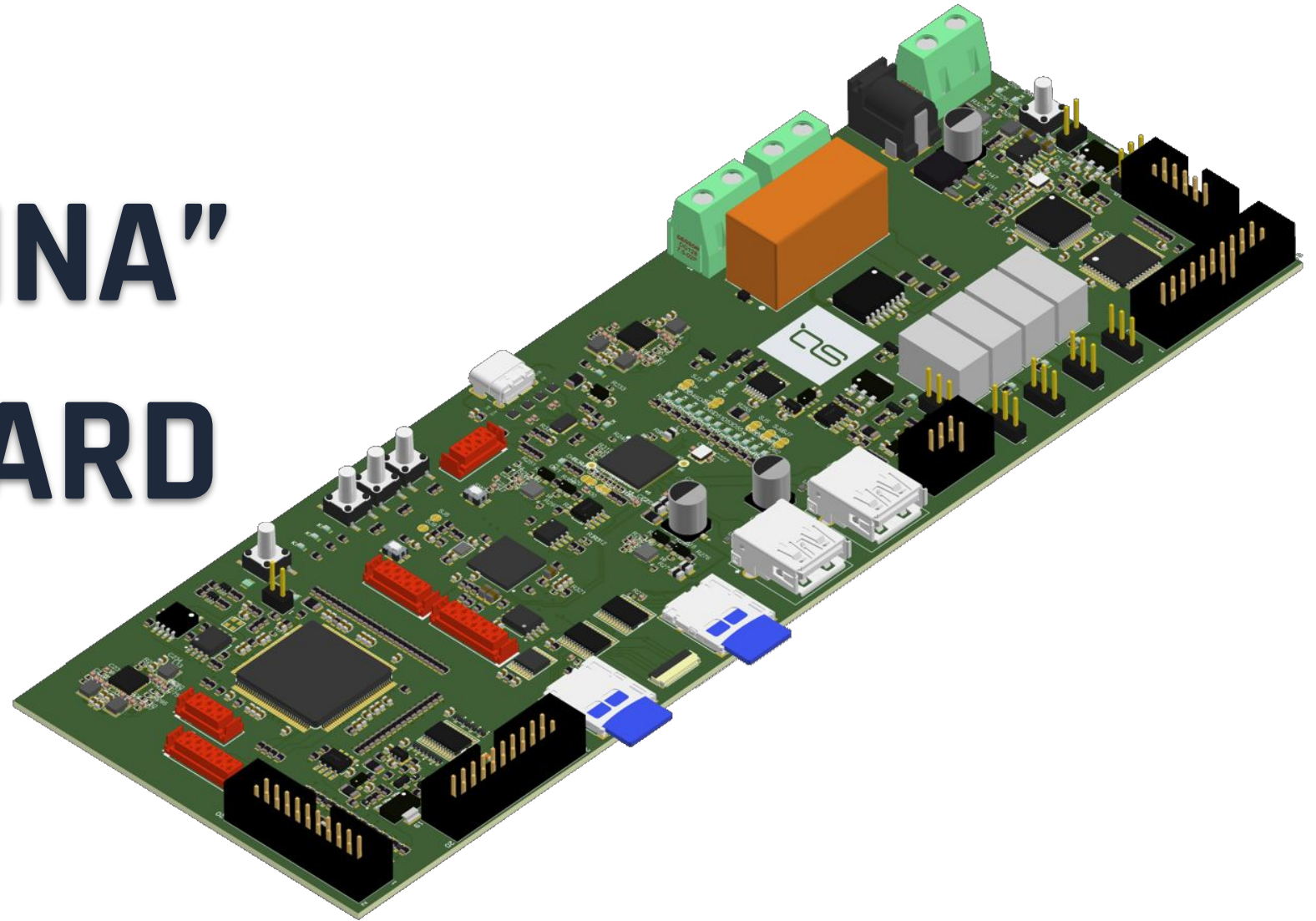


“BIANCHINA” TEST BOARD

Specifications v. 1.0



| BOARD ARCHITECTURE



POWER & CONTROL

DUT Power Supply

Virtual Switches

Super Speed USB Hub



CORE LOGIC

FX3 Controller (ARM9)

JTAG Programmer (FT2232H)

SD-Card Loader & Swap



ANALYSIS

Logic Analyzer

Pattern Generator

Bus Protocol Analyzers

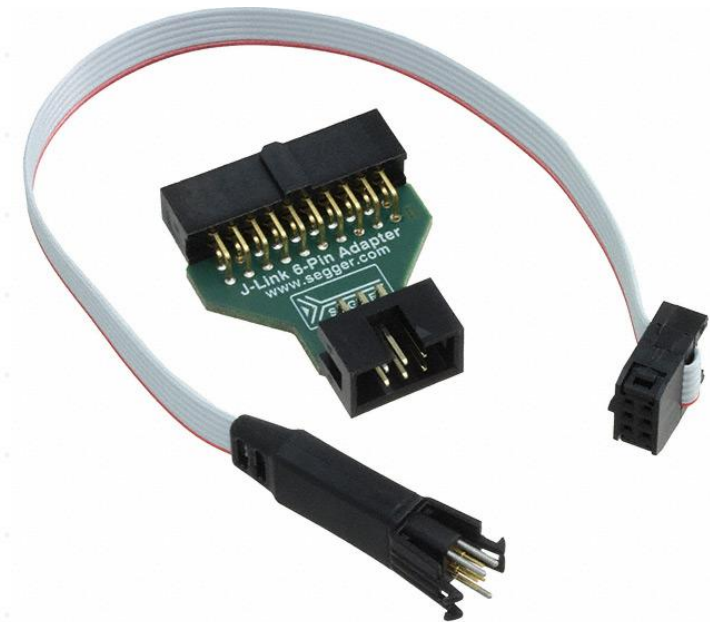
| POWER SUPPLY UNIT

- **Input:** Type-C socket (No data), PD/QC 3.0 compliant. Supports wall phone quick chargers.
- **USB Outputs:** Feeds two USB 3.0 downstream ports (900 mA each).
- **Fixed Output:** Provides one fixed +5V via terminal block.
- **Variable Output:** 1.2V to 3.3V (I2C controlled) for IO interfaces.
- **Measurement:** Integrated circuit measures AC/DC values (up to 250Vac / 350Vdc) via FX3.



| JTAG PROGRAMMER

- **Core:** Based on FT2232H with high-speed USB 2.0 host interface.
- **Compatibility:** Open hardware design. Works with OpenOCD, urJTAG (compatible with 'jtagkey', 'KT-link').
- **Flexibility:** Reprogrammable buffer supports ARM SWD, FPGA, CPLD, and Flash programming.
- **Interfaces:** Can function as I2C, SPI, or Bit-bang interface.
- **Voltage:** Accepts target supply from 1.5V to 5V.

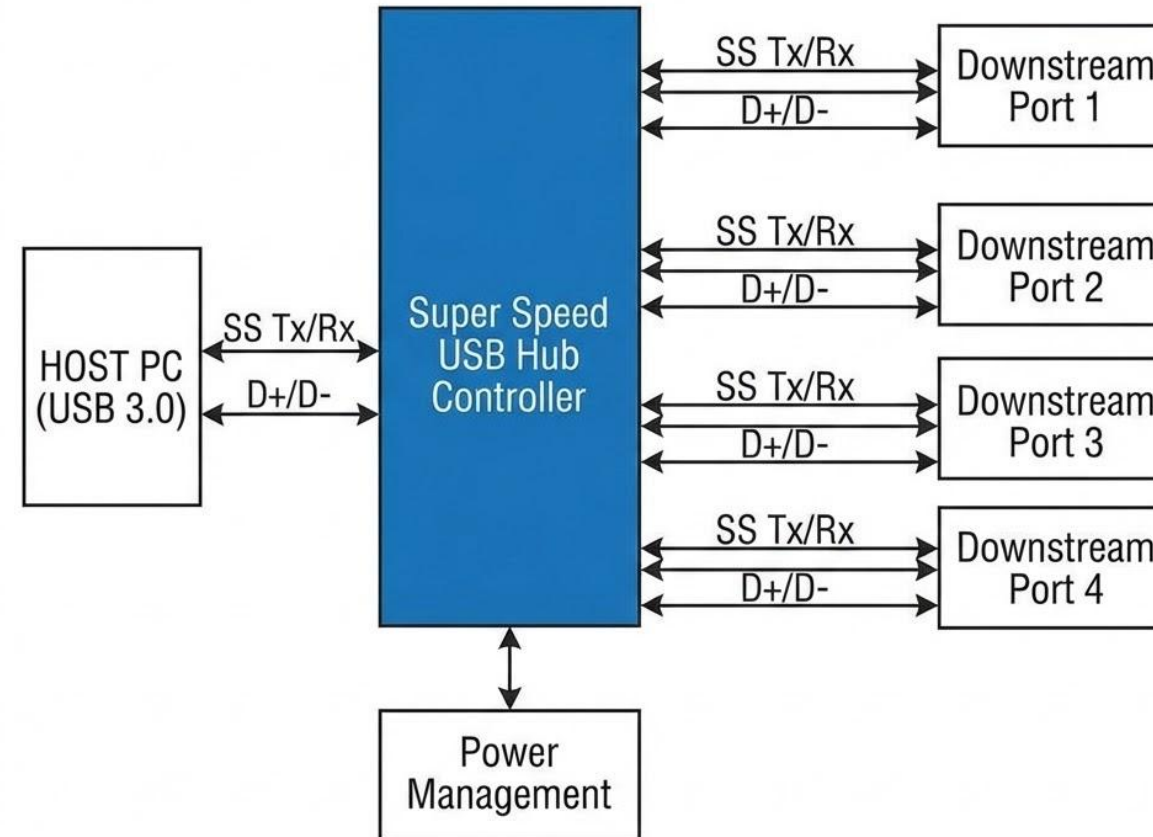


| SUPER SPEED USB HUB

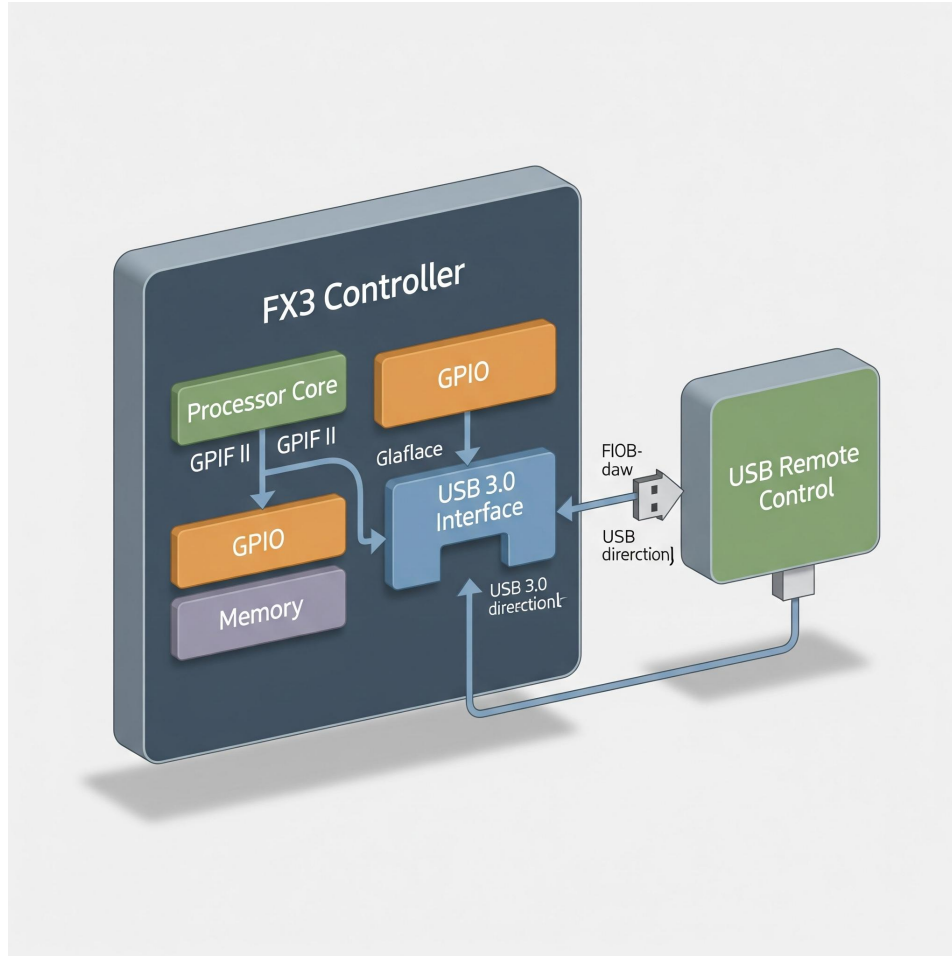
Connects the host PC to all board blocks via a Type-C plug connector.

- **User Ports (1 & 2):** Two standard USB-A sockets (USB 2.0/3.0) freely available for user devices.
- **Management Port (3):** Directly connected to FX3 processor for data management, SD loading, and terminal interface.
- **Debug Port (4):** Connected to JTAG programmer and RS232 interface via FT2232H.

SUPER SPEED USB HUB BLOCK DIAGRAM



VIRTUAL SWITCHES



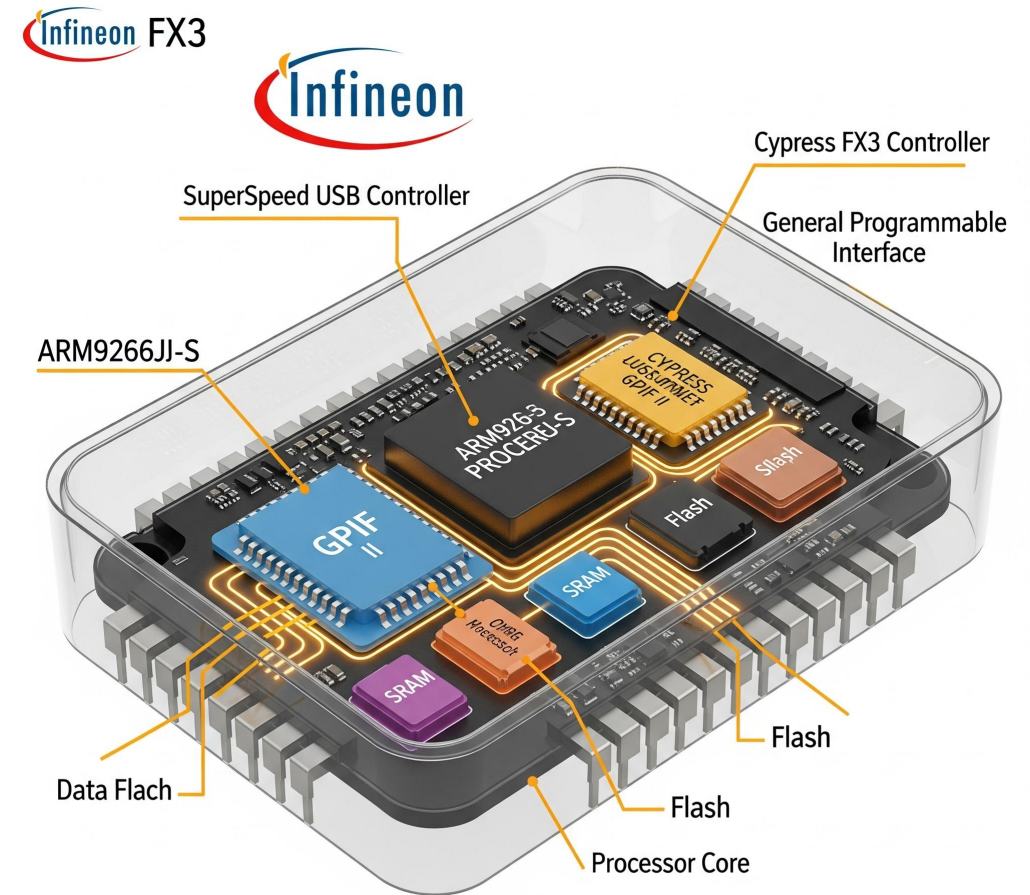
REMOTE CONFIGURATION CONTROL

This block allows for remote activation of switches and jumpers, essential for automated reset and configuration tasks.

- **Composition:** Four DTSP relays.
- **Control:** Managed by the FX3 processor via I2C interface.
- **Connectivity:** Each relay connects to a 3-pin contact header.

FX3 CONTROLLER

- **Processor:** 32-bit ARM926EJ-S with integrated USB 3.0/2.0 PHYs.
- **Memory:** 512 KB or 256 KB on-chip SRAM.
- **Connectivity:** GPIF II interface for glueless connection to SRAM and other interfaces.
- **Performance:** DMA-centric architecture enables 375-MBps direct data transfer.
- **Storage:** Integrated controller supports two mass storage devices (SD 3.0, eMMC 4.41).

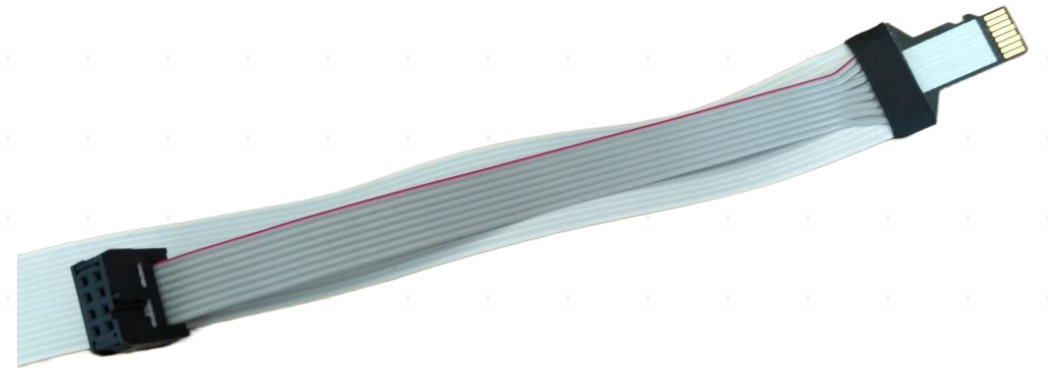
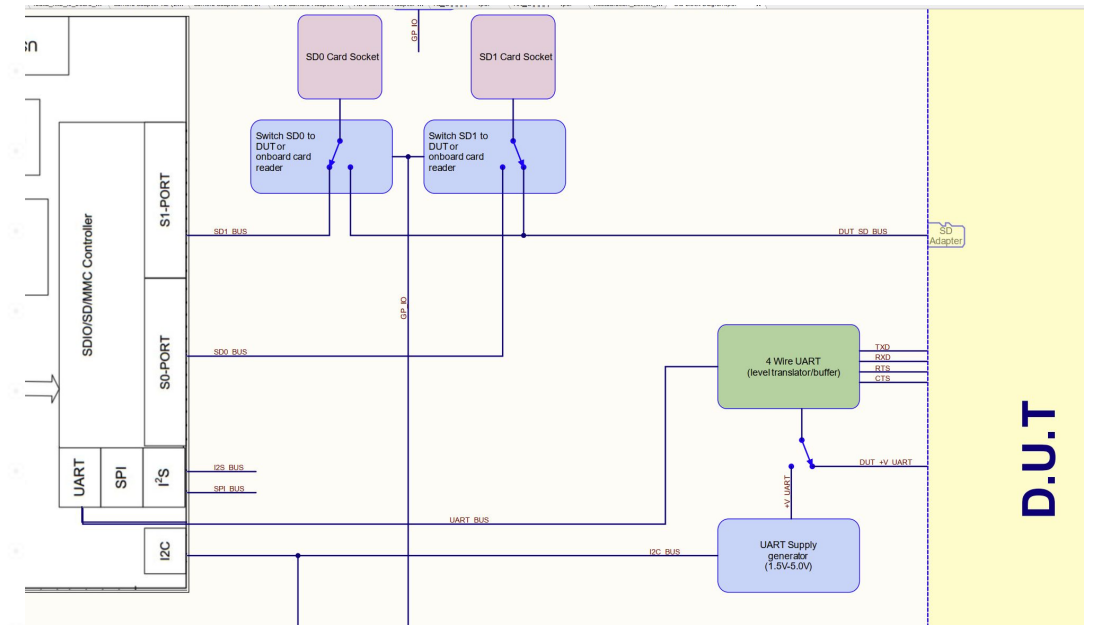


| SD-CARD LOADER & SWAP

SEAMLESS DATA EXCHANGE

Designed to provide the DUT with a programmed SD-card while simultaneously loading data onto a second card for immediate swapping.

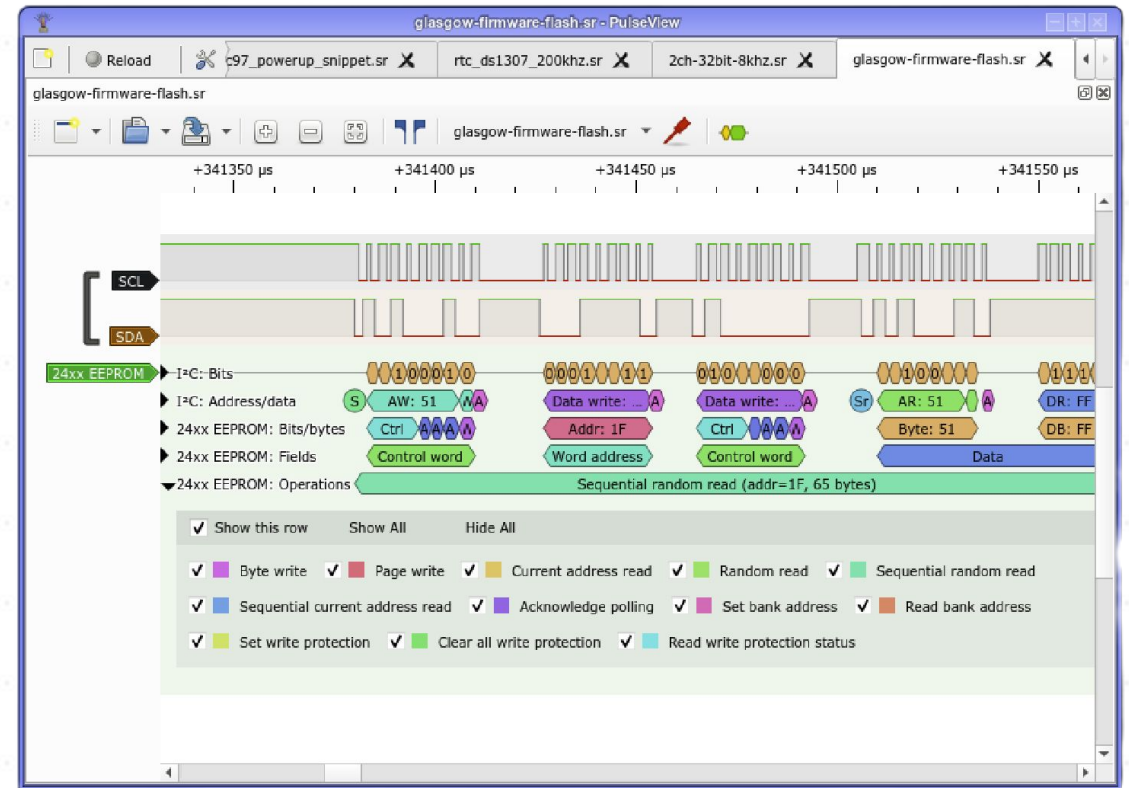
- **Hardware:** Two SD-card sockets and an output flat cable replicating SD contacts.
- **Multiplexing:** High-speed circuit connects cards alternatively to FX3 or DUT.
- **Loading:** FX3 loads data via High Speed USB when connected.



LOGIC ANALYZER SPECIFICATIONS

A combined logic analyzer and pattern generator for embedded development.

- **Inputs:** 10 high-speed channels (DIN0..9) @ 100MSps.
- **Digital I/O:** 8 channels (DIO0..7) @ 50MSps.
Pattern/Protocol gen.
- **Voltage Levels:** User programmable LVCMOS (1.2V to 3.3V), 5V compatible.
- **Bandwidth:** 100 MHz signal input bandwidth.



| ADVANCED ANALYSIS FEATURES



TRIGGERS

Multiple trigger options including pin change and bus pattern detection.



PROTOCOLS

Built-in Digital Bus Analyzers for SPI, I²C, UART, I2S, CAN, and Parallel.



POWER

USB bus powered. Integrated programmable supply (1.2V...3.3V/100mA) for I/O buffers.

TECHNICAL SUMMARY

Feature	Specification	Notes
Power Input	USB Type-C (PD/QC 3.0)	No data transfer on input
Logic Analyzer	100 MSps Input / 50 MSps I/O	100MHz Bandwidth
Processor	ARM926EJ-S (FX3)	512/256 KB SRAM
JTAG	FT2232H Based	OpenOCD / urJTAG Compatible
Voltage Levels	1.2V to 3.3V Variable	5V Tolerant Inputs

Q & A

Thank you for your
attention.